

Sofi Jeffrey

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Education

Bucknell University

B.S. in Chemical Engineering; B.A. in Mathematics

Lewisburg, PA

Expected May 2028

- **GPA: 3.4/4.0**

- **Relevant Coursework:** Chemical Engineering Mathematical Modeling; Food Science and Technology; Heat and Mass Transfer; Reaction Engineering; Engineering Thermodynamics; Separation Processes

Work Experience

Boston Beer Co. – Dogfish Head Craft Brewery

Sensory Co-op

Milton, DE

Jan 2026 – Jul 2026

- Developed mass-transfer model to estimate dimethyl sulfide (DMS) levels in undersized reclaimed CO₂ skid
- Analyzed historical brewing data using principal component analysis (PCA) in Python
- Facilitated internal product testing of 700+ samples

Anheuser-Busch – Research Pilot Brewery

Research Pilot Brewery Co-op

St. Louis, MO

Jun 2025 – Dec 2025

- Managed production and packaging of 60+ barrels (bbl) of product per week
- Collected and interpreted laboratory data for newly developed products
- Revised standard operating procedures (SOPs) for small-batch mixing and pilot-scale fermentation

Research and Modeling Projects

Kinetic Model for Single-Yeast Sour Beer Fermentation

Spring 2023 – Spring 2025

- Developed a kinetic model to represent metabolic retooling during sour beer fermentation
- Conducted an experimental fermentation with continuous gravity and pH measurements
- Calibrated model parameters to capture fermentation behavior across multiple datasets

The Envelopes of Emma Kunz

Published 2025

- Co-authored a peer-reviewed article in *The Mathematical Intelligencer*
- Derived and applied mathematical envelope methods to compute algebraic curves in three artworks

Mass Transfer in a Mr. Coffee

Spring 2025

- Applied engineering assumptions and simplifications to construct math model from first principles
- Modeled flow and mass transfer, including diffusion, during coffee brewing using COMSOL Multiphysics

Matrices with Multiple Crouzeix Extremal Functions

Summer 2023 – Spring 2026

- Authored an operator theory manuscript under review at the *Rose-Hulman Undergraduate Mathematics Journal*
- Generalized and proved patterns involving extremal functions for $n \times n$ matrices

Teaching and Leadership

Bucknell University – Teaching and Learning Center

Quantitative Support Assistant

Lewisburg, PA

Aug 2023 – Present

- Led weekly supplemental calculus lessons for 20+ students
- Hosted Calculus I–III study sessions to improve students' learning outcomes and academic success
- Mentored and trained new and returning staff in methods and techniques to improve program efficiency

Mathematical Association of America – Bucknell Chapter

Vice President

Lewisburg, PA

2024–2025 Academic Year

- Planned, organized, and led campus events for the promotion of mathematics and the mathematical community
- Organized and hosted the first integration bee at Bucknell University

Skills

Programming and Computation: MATLAB, Python, Mathematica, Java, VBA

Engineering Software: COMSOL Multiphysics, Aspen HYSYS, Aspen Properties

Methods and Tools: Differential equations, parameter estimation, PCA, Microsoft Excel (VBA), LaTeX

Interpersonal: Experienced in teaching, mentoring, technical communication, and group facilitation; conversational French